Disagreement Is Not the Signal: Audit-Panel Uncertainty Does Not Recover Critic Blind Spots on Selected Agent Trajectories

Working draft v0.2 — SilentFail-Bench, pilot and Amendments 1–4

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Abstract

When an LLM agent commits the best of n attempts under a critic, selection concentrates the critic's mistakes: the committed trajectory is exactly where its confidence and its errors coincide. A natural safeguard is an *audit panel* of reward models disjoint from the selecting critic, whose disagreement u might flag failures the critic's own score c cannot. We pre-registered and ran this test on SWE-bench Verified with official test-suite gold. The premise held: 35% of committed, agent-declared-complete trajectories failed their hidden tests. The safeguard, as proposed, did not. Across input windows of 4k, 8k and 16k tokens, six chat-preference reward models from 1B to 8B including the proposal's named critic, a code-trained reward model, two selection regimes with genuine best-of-4 selection, and a single-family panel of LLM judges shown the patch, panel disagreement on a single window never rose above chance (AUROC(u) = 0.44–0.54) and never beat the selecting critic's own score. Two integers read off the diff and the transcript — lines added (0.665) and step count (0.657) — outrank every reward model at 4k, and no critic adds out-of-fold value beyond them. What does carry signal is individual critics given enough of the solution: a 1B reward model at 16k reaches 0.71 and LLM judges with the patch in view 0.72–0.75, yet their *disagreement* stays at 0.53–0.64. The one panel statistic that clears the structural bar — the maximum disagreement over pruned, overlapping windows, 0.686 — does so on twelve positives with a permutation p of 0.04 that rises to 0.21 within repository. A test–retest experiment withdraws our earlier mechanism claim: the critics are stable to surface form ($r \geq 0.98$) and respond to window content; the failure is in what disagreement measures, not in critic noise. We report the instrument, the pre-registration trail with every deviation, and what a decisive study would need.

1 Introduction

Best-of-n selection under a learned critic is now a default in agent scaffolds: generate several trajectories, score each with a reward model, commit the argmax. The winner's curse makes this dangerous in a specific way. Among candidates, the one with the highest critic score is disproportionately the one whose flaws the critic cannot see; selection is a filter for the critic's blind spots. A committed trajectory that fails its task while carrying a high score is a *silent failure*.

The obvious safeguard is a second opinion — or several. If a panel of reward models, disjoint from the selecting critic, disagrees sharply about a committed trajectory, that disagreement might carry information the selecting critic has already exhausted. The proposal behind this work stated the hypothesis as a falsifiable, selection-conditional estimand and committed in advance to publishing the

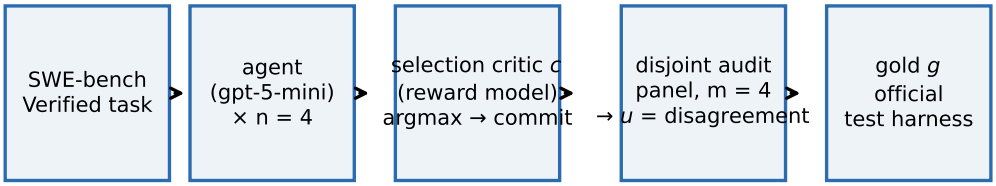


Figure 1: The design. c is the selecting critic's scalar score of a whole trajectory; best-of-n commits the argmax. A disjoint panel of m reward models produces u, the sample standard deviation of their rank-normalized scores (rank norms fit on training repositories only). Gold g is the official SWE-bench harness verdict on the committed patch.

null.

This paper reports that result. The contribution is fourfold. (1) An instrument: real executed gold on SWE-bench Verified, genuine reward-model selection and audit, sandboxed and reproducible, at $3.65 per 164 trajectories. (2) A pre-registered test of the hypothesis that fails for single-window disagreement across every configuration a reviewer would reasonably ask for, and a windowed variant that clears the bar only marginally. (3) A dissociation, established by test–retest: individual critics carry a moderate, content-driven signal about silent failure once they see enough of the solution, but their disagreement does not. (4) A baseline any future audit statistic must beat: patch size, and the finding that no critic tested adds out-of-fold value beyond it.

2 Setup and estimand

Unit and gold. The unit is one committed trajectory τ. Gold $g(\tau) \in \{0, 1\}$ is trajectory-level task success under the official `swebench` 5.x harness; failed tests do not imply every step was wrong, and we make no step-level claims. Positive class throughout is failure (g=0).

Selection and audit. c_{sel} scores the full committed trajectory text; best-of-n ranks n full candidates by this score and commits the top-1. The audit panel $\{c_1, \ldots, c_m\}$ satisfies $c_{\text{sel}} \notin \{c_i\}$ at checkpoint level; u is their disagreement.

Primary population. Committed trajectories whose agent exited with a submission and a non-empty patch — the agent believed it was done. Within it, silent failures are g=0.

Metrics. M1: AUROC(u) for silent failure. M2: AUROC($-c$). M3 = M1 − M2 with a repository-clustered bootstrap (4,000 resamples, groups resampled with the fitted scores held fixed). A held-out logistic ΔAUROC was pre-specified as confirmatory only when ≥ 10 positives fall in the test split; with 41 instances it never did, and we report raw-score AUROCs as registered.

Pre-registration. Population, positive class, metrics, predictions and a stopping rule ("fewer than ten positives yields no conclusion in either direction") were committed to the repository before any gold existed (`PREREGISTRATION_SWEBENCH_PILOT.md`). Each later analysis was added as a dated amendment before its data existed. Every deviation is declared in §6.

3 Instrument, and what it took to trust it

The pilot runs on a laptop and free GPU tiers: mini-swe-agent 2.4.6 driving `gpt-5-mini` through the OpenAI API ($0.02 per trajectory with prompt caching), SWE-bench images under Rosetta in a Colima VM on Apple Silicon, reward models on Metal or on Kaggle's free T4s, and the official harness for gold.

We record three instrument failures because each produced a *uniform* failure rate that looked like a finding, and each was caught only by an oracle control. (i) A sandbox flag hid `numpy` from the gold harness, scoring every trajectory as a failure; caught because the benchmark's reference solutions are run through the harness first. (ii) An LLM-as-judge selecting critic emitted only three distinct values over 640 candidates, so best-of-16 silently became first-draw: 39 of 40 committed trajectories were identical to the n=4 run. Selection had never happened in the earlier pilots. (iii) On a fresh VM the harness's image pulls failed on an architecture manifest, yielding 41/41 failures in 499 seconds; caught by the harness's own error count. The rule we adopted: a rate of exactly 0% or 100% with suspiciously fast timing is an instrument, not a result.

Two statistical defects in our own code were found and fixed before any real run: a power function that reported ≈0.40 "power" under the true null, and repository grouping that truncated hyphenated repository names, corrupting the held-out split. Both have regression tests.

4 Results

4.1 The premise holds: selection concentrates silent failures

Fifty instances were sampled with a fixed seed; generation completed for 41 (the deviation is explained in §6). Of 164 candidates, 138 submitted a patch (17.7% hit the 40-step limit or submitted nothing). Selection by `Skywork-Reward-V2-Qwen3-4B` committed 34 usable trajectories; the harness resolved 22 and failed 12, with zero errors.

35.3% [21.5, 52.1] of committed, agent-declared-complete trajectories were silent failures. On HumanEval+ the same pipeline yielded 8% at any selection pressure; SWE-bench supplies the phenomenon in quantity. Selection was real (Fig. 2, top left), and $\mathrm{corr}(c, u) = 0.11$, so u was near-orthogonal to c — the precondition the design wants.

4.2 The safeguard does not: the ladder of nulls

Table 1 and Fig. 3 are the core of the paper. The primary analysis — four chat-preference reward models auditing a 4,096-token prefix — gave $\mathrm{AUROC}(u) = 0.508$. Every subsequent amendment was declared before its data existed and asked whether some specific limitation of that primary was hiding the signal:

- *The critics never saw the patch.* Scoring the final diff plus the last 3,000 tokens: u fell to 0.458.

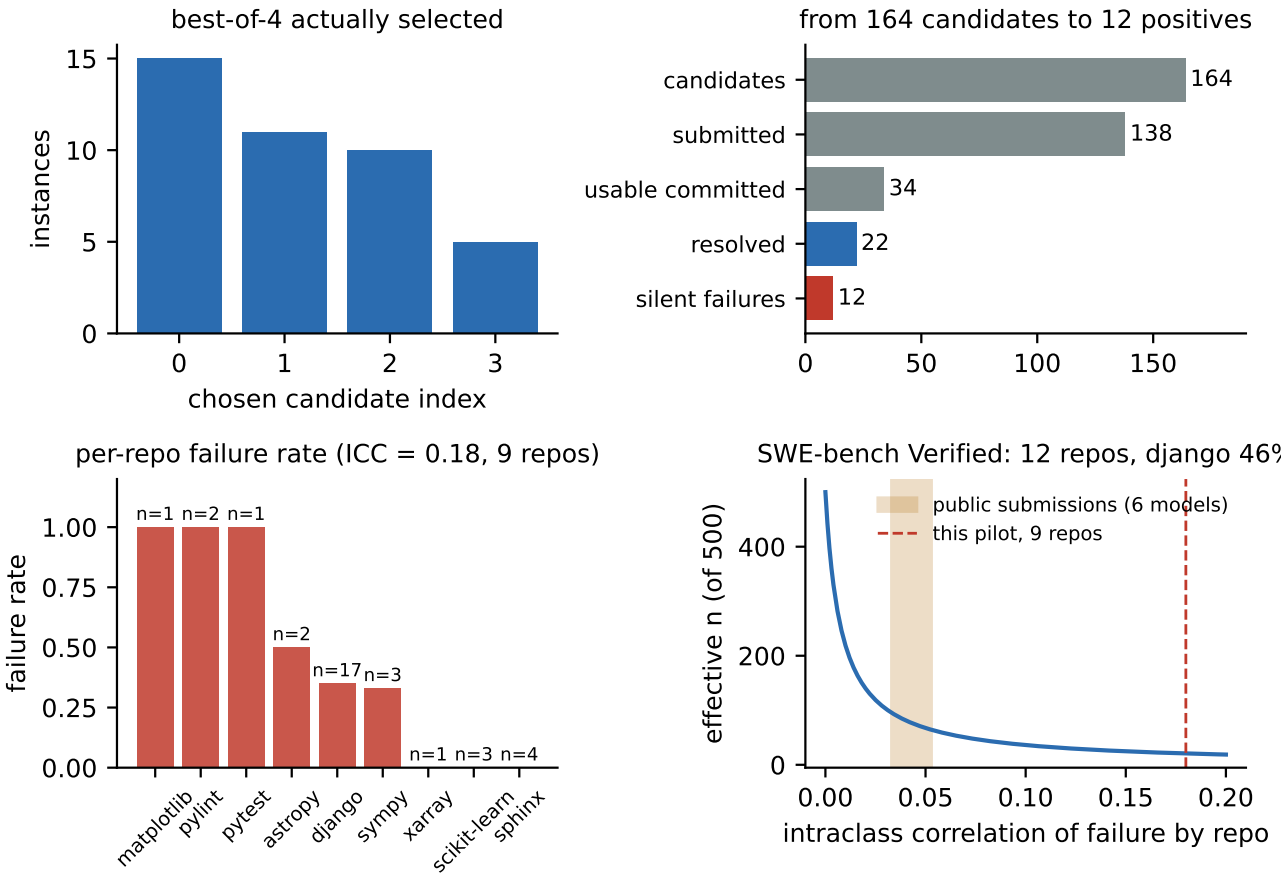


Figure 2: The pilot. Top left: best-of-4 with a true reward-model critic genuinely selected (chosen index spread across all four slots; an earlier LLM-judge critic had collapsed to index 0). Top right: from 164 candidates to 12 silent failures. Bottom left: per-repository failure rates, 9 repositories, ICC 0.18. Bottom right: effective sample size on SWE-bench Verified as a function of repository ICC; the public estimate from six leaderboard submissions is 0.033–0.053.

- *The critics were too small, or the wrong ones.* Six chat-preference RMs from 1B to 8B, including the proposal's named critic `Skywork-Reward-V2-Llama-3.1-8B`, each scored with the patch in view: as individual detectors of the 12 known silent failures they scored 0.45–0.64, five of six at or below chance; the panel gave 0.494.
- *Selection was by the wrong critic.* Re-selecting with the named 8B critic changed 20 of 34 committed trajectories; graded, the failure rate was again 35%, and u was 0.445.
- *Chat-preference training is the problem.* `AceCodeRM-7B`, trained on code with execution-feedback preferences, scored **0.405** as a detector — below chance — and adding it to the panel left u at 0.473. As a selector it changed 24 of 34 trajectories; u on that set was 0.542, its own score 0.472.
- *The context was too short.* Re-scored on a 24 GB GPU at a true 16k tokens (every row at the declared length for the six chat RMs; the code-trained model fell back to 4k on 153/164 rows), five of six chat RMs improved as individual detectors — the 1B Skywork model to 0.712 — with paired 4k→16k gains of +0.06 to +0.12 whose intervals all include zero. The panel's disagreement at 16k was 0.439, the lowest of any row. An 8k point sits between and is not monotone.

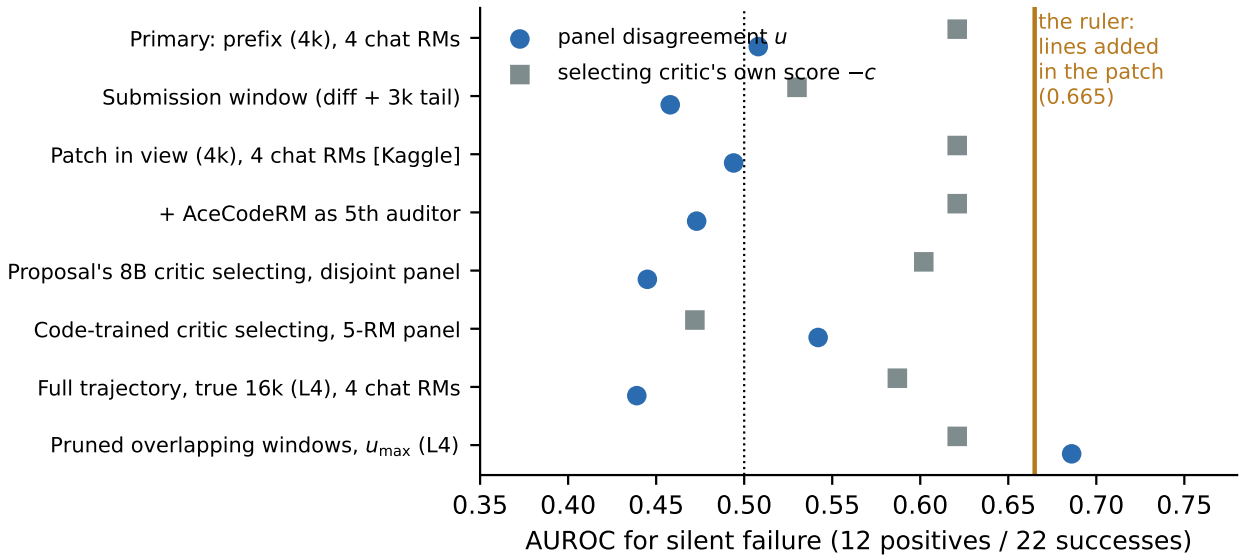


Figure 3: Every pre-registered configuration on the same 34 trajectories and 12 positives. Circles: panel disagreement u. Squares: the selecting critic's own score. Nothing crosses the ruler.

- *Reward models are the wrong class.* Three LLM judges (`gpt-5`, `gpt-5-mini`, `gpt-5-nano`; one vendor, hence a single-family panel) rated the probability that the patch resolves the issue. Shown the problem and the diff, the two larger judges reached 0.72 and 0.75 — above the structural bar — but their disagreement was 0.528. Adding the last 6k tokens of the transcript, including the agent's own test runs, did not help (best judge −0.03; disagreement 0.640). `gpt-5-mini` is also the agent, a self-evaluation confound; `gpt-5` is not, and scored 0.72.

Every single-window prediction failed. The intervals on M3 are wide — twelve positives — and a moderate positive effect is not excluded by any one row; the rows are also re-measurements of the same twelve positives, not independent replications. What they establish jointly is narrower than "no detector": under every variation of input length, size, identity, training signal, critic class and selection regime, *the disagreement of a panel on one window of a committed trajectory* carries no information about its hidden failure that the selecting critic's own score does not.

4.3 Windowed auditing: the one row that clears the bar

Amendment 3, pre-registered before any windowed score existed, made one structural change: collapse tool outputs longer than 24 lines to their first and last 8, then score overlapping 3,500-token windows (25% overlap, problem statement prefixed, final window anchored at the patch) and take the panel's maximum disagreement. All three registered predictions passed: AUROC($u_{\max}$) = 0.686 against a bar of 0.665; a gain of +0.18 over the prefix panel against a bar of +0.10; and the named 8B critic's mean-window score, 0.652, against 0.634.

We report this as the study's strongest positive signal and immediately bound it. A free permutation of labels gives $p = 0.04$; permuting within repository, which removes the contribution of repositories that both fail more and disagree more, gives $p = 0.21$. Leaving out any one of the 34 instances moves the estimate between 0.657 and 0.731 — across the bar. The paired gain over the selecting critic is +0.06 [−0.11, +0.33]. Pruning removed 69% of tokens, so most trajectories fit in one or two windows

Configuration	M1 AUROC(u)	M2 AUROC($-c$)	M3	95% CI
Primary: prefix (4k tok), 4 chat RMs	0.508	0.621	−0.114	[−0.37, +0.22]
robustness, m=3 (family-disjoint)	0.545	0.621	−0.076	[−0.39, +0.23]
Submission window: diff + last 3k tok	0.458	0.530	−0.072	[−0.35, +0.13]
Patch in view (4k), 4 chat RMs	0.494	0.621	−0.127	[−0.28, +0.11]
+ AceCodeRM-7B as 5th auditor	0.473	0.621	−0.148	[−0.30, +0.10]
8B named critic selecting; disjoint panel	0.445	0.602	−0.157	[−0.56, +0.14]
Code-trained critic selecting; 5-RM panel	0.542	0.472	+0.069	[−0.21, +0.45]
Full trajectory, true 16k tok (L4), 4 chat RMs	0.439	0.587	−0.182	[−0.38, +0.17]
LLM judges (gpt-5 family), patch in view, u_{LLM}	0.528	0.621	−0.093	[−0.30, +0.10]
execution-aware (+ last 6k tok of transcript)	0.640	0.621	+0.019	[−0.28, +0.16]
Pruned overlapping windows, $u_{\max}$ (L4)	**0.686**	0.621	+0.064	[−0.11, +0.33]

Table 1: The pre-registered primary and all amendments. Same 34 trajectories; 12 positives in every row except the two re-selections (12 and 11). M2 is the selecting critic's own score at the row's input where it was re-scored, otherwise the primary c. CIs are repository-clustered bootstraps on the paired difference. Only the last row clears the structural bar of 0.665 (§4.3).

(median two); "multi-window" here mostly means "the pruned trajectory, plus one overlapping cut". In 43% of multi-window rows the maximum fell on a window other than the last, so the middle of the trajectory contributes something. $u_{\max}$ is also a maximum of noisy statistics and is biased upward under the null. A reader who takes 0.686 as a detector is reading more than twelve positives can support; a reader who dismisses it is ignoring the only pre-registered prediction in the study that came true.

4.4 The critics carry a signal; their disagreement does not

Two further facts sharpen the picture. First, single critics given enough of the solution reach 0.71–0.75 (the 1B reward model at 16k; the two larger LLM judges with the patch in view), while every disagreement statistic built on the same scores lands at 0.44–0.64. Disagreement is not extracting what the critics know. Second, a pre-registered incremental-value test (Amendment 4a) asked whether u or c adds anything to a structural model of the diff and transcript under leave-one-repository-out evaluation: u adds −0.03 [−0.08, +0.04]; c adds +0.02 [−0.10, +0.21]. We declare one defect in that amendment: the registered "difficulty" control, the fraction of six public leaderboard runs that solved the instance, turned out to predict our silent failures at 0.95 on its own — one of those runs is our own agent model, and the others fail the same instances — so the registered model was at ceiling and uninformative. The numbers above are from the labelled structure-only sensitivity variant; all three variants are in the record.

4.5 A ruler beats the critics

Two integers, read off the diff and the transcript with no model at all, rank silent failures better than every reward model at 4k and every disagreement statistic at any length (Fig. 5): lines added in the patch, 0.665; agent step count, 0.657. Bigger patches and longer trajectories fail more. The paired gap over the primary u is +0.16 [−0.11, +0.36], exploratory, on twelve positives. Individual critics do cross this bar once they see the solution at length (§4.3); the panel's disagreement never does on a single window, and adds nothing to the ruler out of fold. We propose lines-added as the baseline any audit statistic must clear before it is worth a confirmatory run — and note that on SWE-bench Verified, beating it by 0.10 in a paired test has power of about 0.3 at any n, because twelve repositories

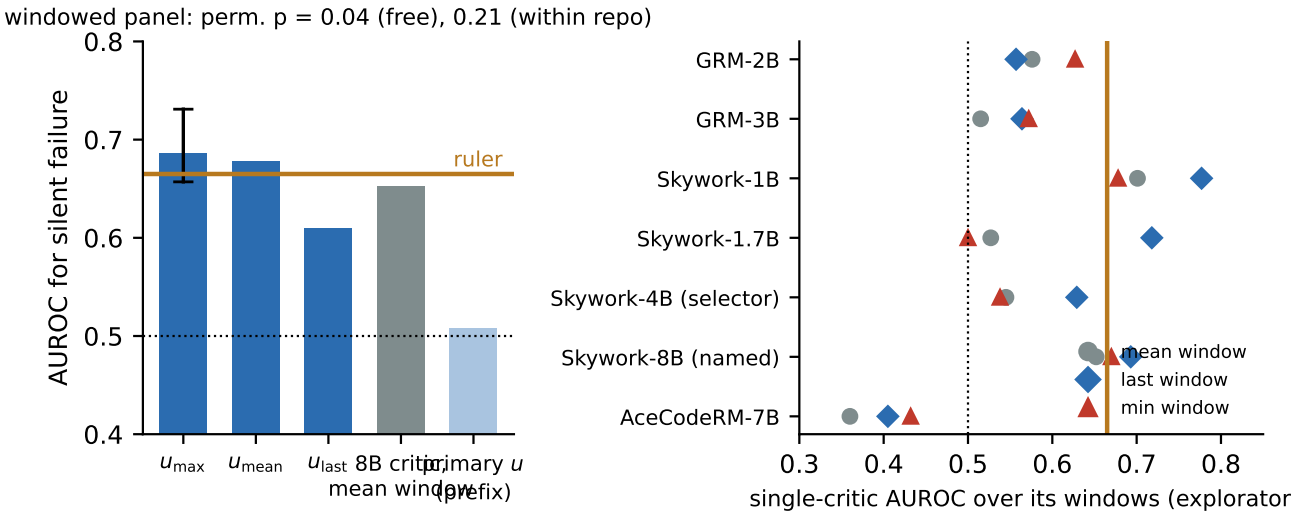


Figure 4: Amendment 3. Left: panel disagreement aggregated over pruned, overlapping windows; whisker on $u_{\max}$ is the leave-one-instance-out range. Right: each critic's own score aggregated over its windows (exploratory; no prediction was registered for these).

cap the effective sample (§6).

5 Mechanism: stable critics, uninformative disagreement

Draft v0.1 of this paper read Fig. 6 — the same reward model scoring the prefix and the patch of one trajectory with correlation 0.03–0.18 — as evidence that these critics "do not hold a judgment" of code correctness. That reading could not distinguish critic noise from windows that legitimately contain different information, and we pre-registered a test to separate them (Amendment 4b, with the reading fixed in advance). The result withdraws the claim. Re-scoring the identical window after a reformatting that changes no information, every critic reproduces its ranking at $r = 0.98$–0.99; after a 128-token shift, at 0.86–0.95; the per-window panel u itself at 0.91 and 0.73. Adjacent windows with 25% overlap correlate at 0.20–0.56, first against last at -0.39 to 0.47. These critics are stable instruments responding to what is in front of them. The prefix score is a judgment of the problem; the patch score is a judgment of the patch; they differ because the inputs do.

The failure is therefore in the statistic, not the critics. A panel's disagreement is large when its members' rank-normalized scores spread out — which they do for reasons that have nothing to do with correctness: different critics are calibrated to different aspects of a window, and the spread is dominated by that between-critic heterogeneity rather than by any shared uncertainty about the solution. The signal the critics carry individually (0.71–0.75 at length) is averaged away, not amplified, by measuring how much they disagree. Windowed $u_{\max}$ recovers some of it, plausibly because the maximum over windows behaves partly like a max over the critics' own low scores; its exploratory single-critic analogues (Fig. 4, right) are consistent with that.

This explains the earlier false positive too. On HumanEval+ with an LLM-as-judge critic, we briefly observed $u = 0.69$ and a "Goodhart signature" of the critic below chance. Both were artifacts of a critic with three output values on three positives; with a real reward model selecting, both vanished. We withdrew the claim in the record before running the SWE-bench pilot, as we withdraw the instability

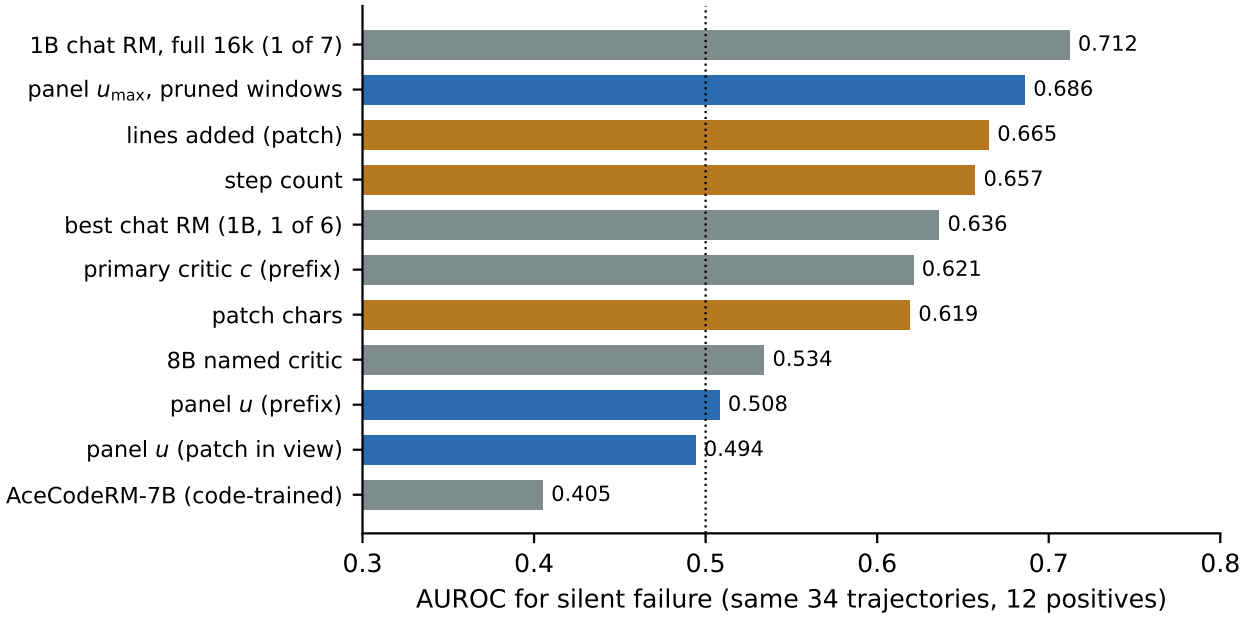


Figure 5: Same 34 trajectories, same 12 positives. Gold bars need no model. Exploratory (declared as such): the covariates were registered as controls, not baselines.

claim here.

6 Limitations and declared deviations

- **Twelve positives.** Every inference is at the pre-registered eligibility floor. CIs are wide; the argument rests on consistency across configurations, not on any one.
- **41 instances, not 50.** Generation stalled at instance 41 when macOS swap exhausted the host disk. The missing nine were determined by list order and the time of the stall, not by the instances.
- **Repository clustering.** SWE-bench Verified is 500 instances over 12 repositories, django 46%. From six public submissions on this scaffold the ICC of failure by repository is 0.033–0.053, giving an effective n of 64–95 of 500 (Fig. 2); on our 9 repositories it is 0.18. *The cross-repository generalization claim in the original proposal is not testable on this benchmark at any n.* We report repository-clustered intervals and make no cross-repository claim.
- **Input length.** The primary scored a 4k prefix; Kaggle's T4s fell back to 4k on most 16k rows (declared per row). The L4 runs scored every row of the six chat RMs at a true 8k and 16k; `AceCodeRM-7B` fell back to 4k on 153/164 rows at both, so its "16k" number is not one. The ladder is 4k/8k/16k, with all paired gains inside their intervals.
- **Critic classes.** Six chat-preference RMs, one code-trained RM, and three LLM judges from one vendor. The 32B code-trained model did not load on a 24 GB GPU (twice). Cross-vendor LLM judges, and a judge that executes tests, were not run; the claim about LLM judges is single-family.
- **Multiple looks.** Roughly twenty configurations were scored against one bar on the same twelve positives. The individual crossings (one reward model of seven at 16k; two judges of three; $u_{\max}$ with $p = 0.21$ within repository) are what one expects from that many looks under a weak true

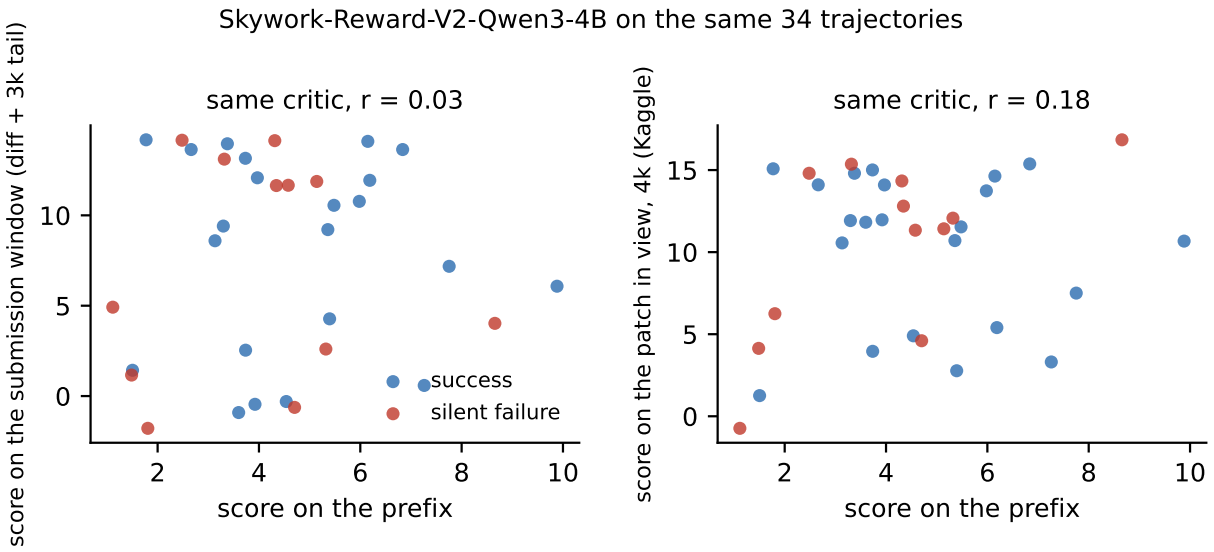


Figure 6: The primary selecting critic's score on the trajectory prefix (x) against its score on two other windows of the same trajectory (y). Same model, same trajectory, nearly unrelated numbers — because the windows contain different things (see Fig. 7).

effect, and are reported as such.

- **One agent, one task family.** gpt-5-mini at 40 steps on Verified. Silent-failure structure may differ for stronger agents or other benchmarks.
- **Windowed auditing is thin.** Pruning left most trajectories at one or two windows; $u_{\max}$ over two windows is not the eight-window design the amendment described, and its sensitivity to single instances is shown in Fig. 4.
- **The difficulty control failed.** Amendment 4a's registered control (public solve rate) is a near-oracle for this agent's failures and was replaced, with the deviation labelled, by a structure-only base.

7 Related work

[TODO: Port citations from the proposal's bibliography: Goodhart/winner's curse under best-of-n (Gao et al. 2023; inference-time reward hacking), reward-model overoptimization, process reward models and Math-Shepherd-style step labels, SWE-bench/Verified and mini-swe-agent, RewardBench and the Skywork/GRM/AceCode reward models, and negative-result and registered-report precedents in ML evaluation.] The nearest prior work studies critic disagreement as uncertainty for *unselected* outputs; to our knowledge no prior evaluation reports a held-out, selection-conditional, beyond-score test with a disjoint audit panel on agent trajectories with executed gold.

8 Conclusion

The premise of critic-audited agents is sound: a third of what a reward-model critic confidently committed on SWE-bench was broken. The proposed safeguard — disagreement among a disjoint panel — did not detect those failures on any single window under any configuration we could pre-register and run, across reward models and LLM judges alike, and it adds nothing to counting the lines

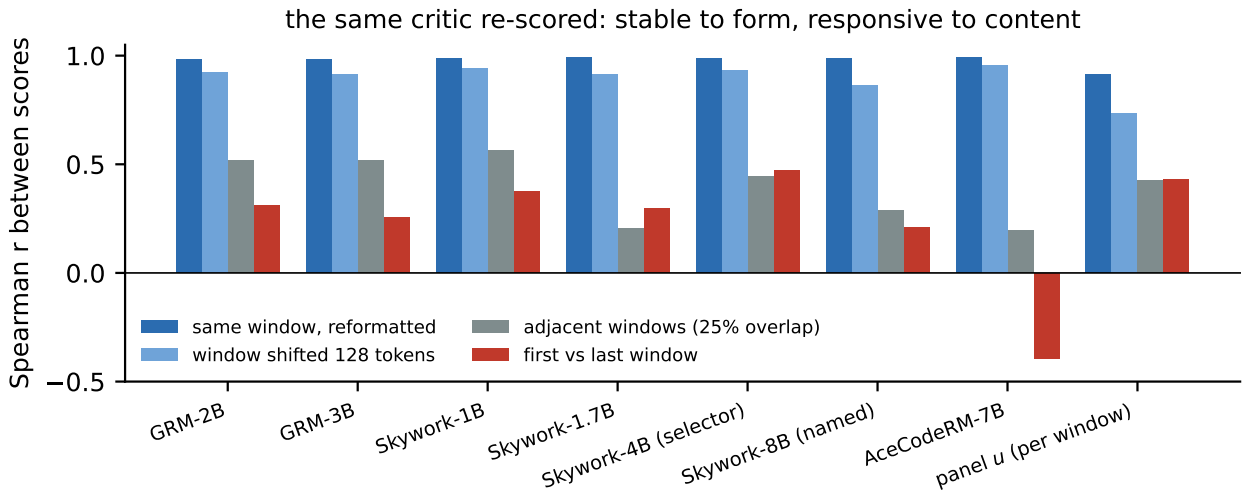


Figure 7: Amendment 4b, test–retest. Each critic re-scored every Amendment-3 window after a content-preserving reformatting (headers re-cased, pruning marker reworded, one neutral line prepended) and after a 128-token shift. Spearman r over 65–73 windows. The same critic on the same content is essentially deterministic; on adjacent or distant windows it is not.

in the patch. The critics themselves are not the problem: they are stable, they respond to content, and given the solution at length they rank silent failures moderately well on their own. What fails is the idea that their *disagreement* is where the missing information lives. The single exception — maximum disagreement over pruned windows, just over the structural bar on twelve positives — is exactly the kind of result that needs forty repositories rather than five hundred more trajectories to settle, and we say so rather than run the confirmatory study the proposal promised. Anyone building an audit panel from off-the-shelf critics should first check whether its disagreement beats counting the lines in the patch. Ours did not; its members, read directly, sometimes did.

A Chronology and cost

Stage	What	Cost
Freezes, defects, calibrated power	18 regression tests	$0
HumanEval+ pilots 1–3b (instrument validation)	40 tasks, $n \in \{4, 16\}$, 2 judges → 4 RMs → RM selection	$0
SWE-bench infrastructure without admin rights	Colima + Rosetta + swebench 5.x; three oracle controls	$0
SWE-bench pilot	164 gpt-5-mini candidates, 41 instances, official gold	$3.65
Amendment 1 (submission window)	local, 5 RMs	$0
Amendment 2 (Kaggle 2×T4)	7 critics, 164 candidates, two re-selections graded	$0
Amendments 2 (true 16k) and 3 (Algoverse L4)	7 critics × 164 candidates; 7 critics × 34 windowed rows	$0
Amendments 4a, 4b, 4d (local; L4 stage 5)	incremental-value model; test–retest 7 critics × 3 variants; 8k ladder point	$0
Amendment 4c (OpenAI API)	3 judges × 2 conditions × 34	$1.96

B Reproducibility

Code, pre-registrations with amendments, every scored row, every critic's scores, the harness reports and the research log are in the repository (`UPDATES.md` has one entry per stage: done / learned / changed). All figures are generated by `paper/make_figures.py` from committed JSON.